



The more decimal places a number has, the smaller the number is.

Do you agree with Tiny?
Explain your answer.



O	Tth	Hth	Thth

Use four counters to make a number less than 1

What is the value of each digit in your number?

How many ways can you partition it?

0.454

0.44

0.445

0.345



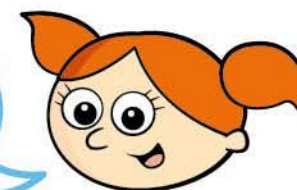
The children are each thinking of a different decimal number.



Amir

My number has
four hundredths.

My number
is the smallest.



Alex



Dora

The sum of
the digits in my
number is 13

The tenths
and hundredths digits
in my number
are different.



Dexter

Match each number to the correct child.



Which is the odd one out?

A $2 + 0.1 + 0.02 + 0.003$

B $1 + 1.1 + 0.02 + 0.003$

C $2 + 1.1 + 0.03$

D $2 + 0.1 + 0.01 + 0.013$

E $2 + 0.1 + 0.023$

Explain your answer.

Create your own question like this for a partner.



O	Tth	Hth	Thth

Use five plain counters to make a number greater than 1

What is the value of each digit in your number?

How many ways can you partition it?

Is the statement always true, sometimes true or never true?



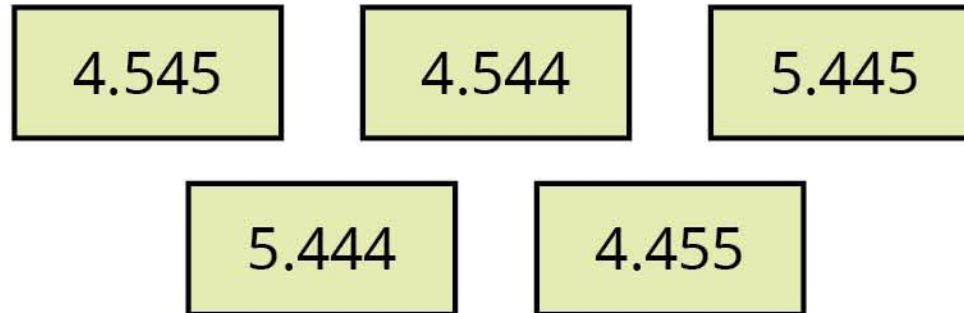
A number with 3 decimal places
is greater than a number with only
1 decimal place.

Explain your answer.





Here are some number cards.



Use each number once only to complete the sentences.

_____ rounded to the nearest tenth is 4.5

_____ rounded to the nearest integer is 4

_____ rounded to the nearest tenth is 5.4

_____ rounded to the nearest hundredth is 5.45

_____ rounded to the nearest hundredth is 4.54



Use the digit cards to make the statements correct.

You may use each card once only.

4

5

6

7

.803 rounded to the nearest integer is 6

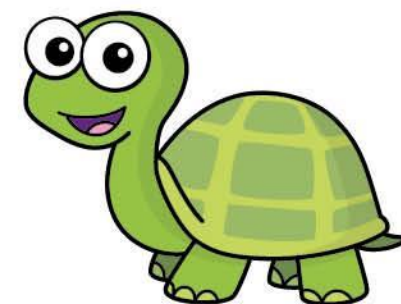
5.9 rounded to the nearest tenth is 6

6. rounded to the nearest integer is 6

.002 rounded to the nearest hundredth is 6

Tiny has represented $16.53 + 5.485$ on a place value chart.

T	O	Tth	Hth	Thths
●	● ● ● ● ● ●	● ● ● ● ●	● ● ●	
● ● ● ● ●	● ● ● ●	● ● ● ● ● ● ● ●	● ● ● ● ●	



What mistake has Tiny made?

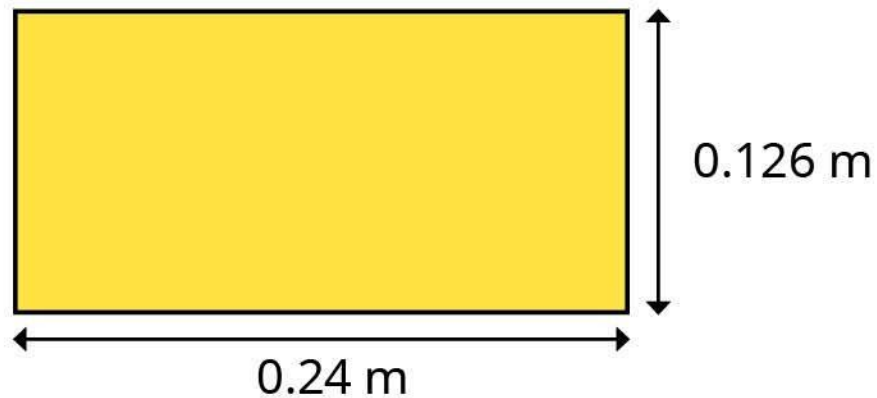
Represent the calculation correctly.

What is the correct answer?

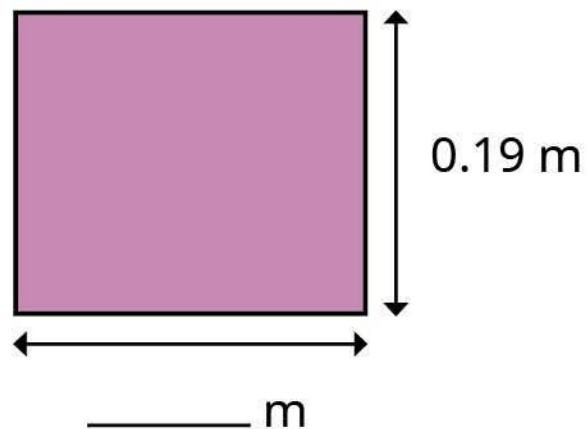




Work out the perimeter of this shape.



This rectangle has a perimeter of 0.866 m.



Work out the missing length.

Tiny is multiplying numbers by 100



When you
multiply by 100, you just
add two zeros to the end
of the number.



Give an example of a calculation where Tiny's method works.

Give an example of a calculation where Tiny's method does **not** work.

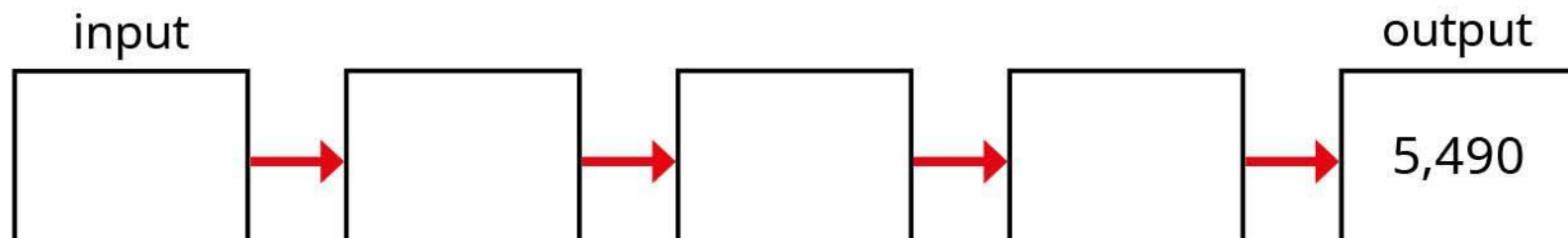
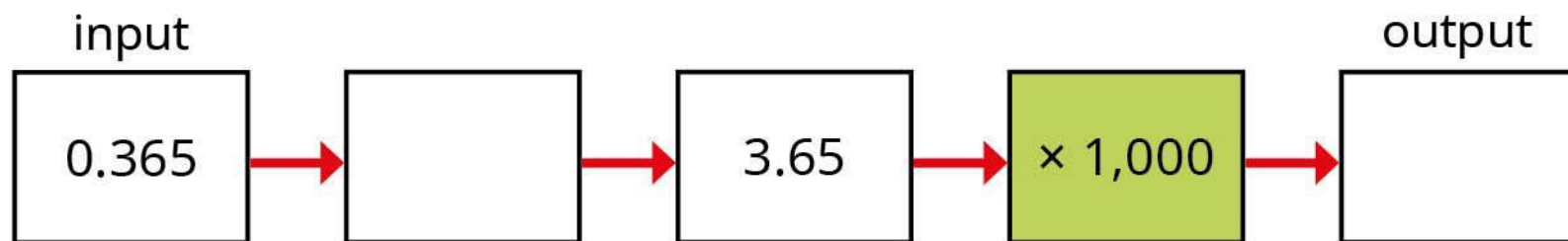
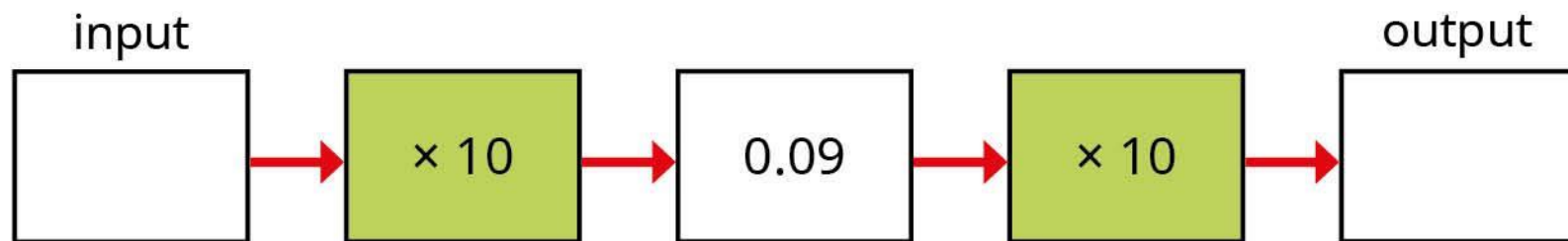
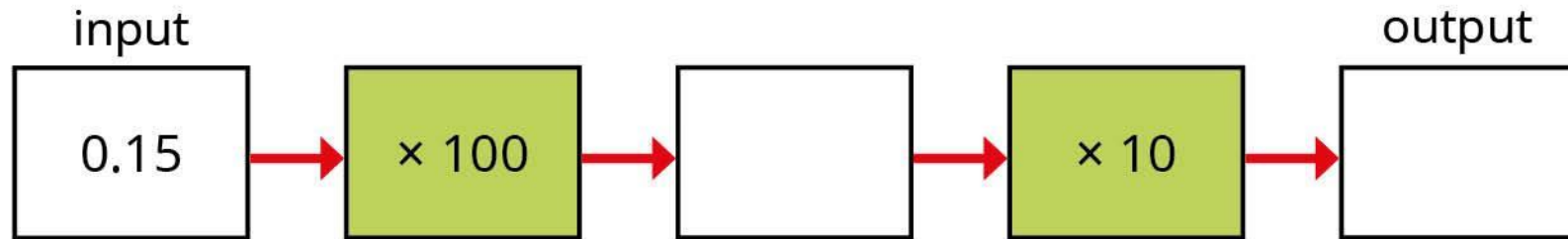
What is a better way to explain how to multiply by 100?

Talk about it with a partner.





Fill in the missing numbers.



Tiny is dividing numbers by 10, 100 and 1,000

When you
divide by 10, 100 or
1,000, you just remove
the zeros.



Do you agree with Tiny?
Explain your answer.



Use the rules and the table to make 70 in as many ways as you can.

- Use a number from column A.
- Use an operation from column B.
- Use a number from column C.

A	B		C
7	×	÷	1
70			10
700			100
7,000			1,000

Is the statement true or false?

Dividing by 1,000 is the same as
dividing by 10 three times.

Explain your answer.





I know that
 $25 \times 4 = 100$,
so $0.25 \times 4 = 0.100$

Do you agree with Tiny?
Explain your answer.



Is the statement always true, sometimes true or never true?

When you multiply a number with 2 decimal places by an integer, the answer will have 2 decimal places.

Explain your answer.





Chocolate eggs can be bought individually, or in packs of 6 or 8



What is the cheapest way for Max to buy 25 chocolate eggs?

How much will he spend?

Tiny uses place value counters to work out $3.27 \div 3$

O	Tth	Hth
1 1 1	0.1 0.1	0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01

I only had two counters in the tenths column, so I moved one of the hundredths so each column could be grouped into 3s.



Explain why Tiny is incorrect.
What is the correct answer?





C is $\frac{1}{4}$ of A

$$B = C + 2$$

Use this information to complete the division.

		0	•	B	B
	A	C	•	^C B	^C 2

Compare methods with a partner.

How did you work it out?

Create your own question like this for someone else to solve.



1.28 kg of sand is shared equally between 4 buckets.

There is 5.12 kg
of sand in each
bucket because
 $1.28 \times 4 = 5.12$



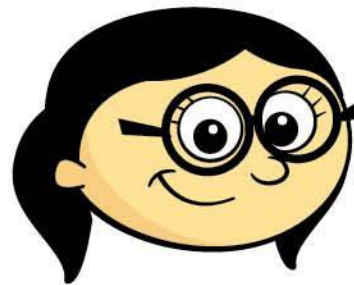
Explain the mistake that Tiny has made.
What is the mass of sand in each bucket?





Annie has some money.

- She gives $\frac{2}{3}$ of her money to charity.
- She then buys three footballs costing £6.45 each.
- Her mum gives her and her two sisters £9.75 to share equally between them.



Now I have
got £10.50

How much money did Annie have to start with?